

CO2-AT-1

⇒ Scenario 1: Phishing Email investigation:

A user receives an email claiming to be from their bank asking them to "verify" their account immediately.

→ Investigation:

The email should be examined carefully before taking any action.

→ Phishing Indicators:1. Verify the sender's Email Address:

→ Check whether the email domain matches the official bank domain.

→ Example:

- fake: support@bank-securityverify.com
- Legitimate: support@bankname.com

→ fake domains often contains extra words, numbers, or spelling mistakes.

2. inspect the links:

→ If the displayed URL is different from the official bank website it is a phishing attempt.

Ex: Displayed: www.bank.com

Actual: http://bank-login.verify-user.xyz

3. Look for Urgent or Threatening language:

→ "Your account will be suspended."

→ verify within 24 hours.

→ immediate action required.

=> Risk Identified:

- Credential
- identity
- financial
- malware
- unauthorized banking transaction

=> Conclusion:

The email shows multiple phishing including a suspicious sender, fake link, and urgent language. It should be treated as a phishing attack and reported immediately.

=> Scenario 2: Social Engineering investigation:

A company notices unusual outbound network traffic shortly after an employee receives a social engineering phone call.

=> Cyber attack Techniques:

- Reveal credentials.
- Install remote access software.
- Disable antivirus
- Execute a malicious program.

=> Possible Attack chain:

- initial contact: IT Support, HelpDesk, Software vendor.
- Social Engineering: Install remote software, login credentials, warnings
- Initial Access: Stolen credentials, Remote desktop software, VPN login
- Execution: The attacker executes malware on the system.
- Command and Control: The infected computer communicates with the attacker's server, causing unusual outbound traffic.

=> Investigation Steps:

- Isolate the computer
- collect Logs.
- Identify malware
- Reset credentials.
- Reset Analyze network Traffic.
- Performance forensic Analysis.

=> Risk Identified:

- Data exfiltration
- Credential theft.
- Remote Access
- malware infection
- Business disruption.

=> Conclusion:

Immediate isolation, forensic investigation and credential reset are required.

=> Scenario 3: Ransomware Investigation:

A corporate server has been infected with ransomware. Trace the attack chain and identify how OSINT could have detected the threat earlier.

=> Attack chain:

- Initial Access
- Execution
- Lateral movement
- Encryption
- Impact

=> Investigation Steps:

- Disconnect the infected server.
- Preserve forensic evidence.
- Identify the ransomware family.
- Review logs:
- Determine patient zero.
- Restore systems from verified backups.

=> Risk Identifiers:

- Data encryption
- Data theft
- Business interruption
- Financial loss
- Regulatory penalties
- Reputation damage.

=> Conclusion:

The ransomware attack likely progressed through initial access → execution → lateral movement → encryption → impact. Proactive monitoring using OSINT, timely patching, strong authentication and user awareness could have reduced the likelihood and impact of the attack.